MIL-C-0081309C(AS)
30 August 1973
SUPERSEDING
MIL-C-0081309B(AS)
12 July 1972
And used in lieu of
MIL-C-81309A
13 June 1968

MILITARY SPECIFICATION

CORROSION PREVENTIVE COMPOUNDS, WATER DISPLACING, ULTRA-THIN FILM

This limited coordination Military Specification has been prepared by the Naval Air Systems Command based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-C-81309A. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 Scope - This specification covers two types and classes of water displacing corrosion preventive ultra-thin film compounds which may be applied by dipping, spray gun, brushing or from gas pressurized containers (see 6.1).

1.2 Classification - The compounds shall be furnished in the following types and classes as specified (see 6.2).

Type I Dry-to-the-touch
  Class 1 - Bulk form (brush, dip, or spray application)
  Class 2 - Pressurized Container (For spray application)

Type II Soft film
  Class 1 - Bulk form (brush, dip, or spray application)
  Class 2 - Pressurized Container (For spray application)

1.3 Relationships of the compounds of this limited coordination Military Specification to the grades of the previous specifications are indicated in 6.6.

FSC 8030

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-F-1421	Fluorocarbon Refrigerants
QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet
QQ-A-250/5	Aluminum Alloy Alclad 2024, Plate and Sheet
QQ-A-671	Anodes, Cadmium
QQ-B-626	Brass, Leaded and Non-leaded, Rods, Shapes, Forgings and Flat Products with Finished Edges (Bars, Flat Wire, and Strips)
QQ-C-576	Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed, or Machined Edges (Plate, Bar, Sheet and Strip)
QQ-M-44	Magnesium Alloy, Plate and Sheet (AZ31B)
QQ-S-766	Steel Plate, Sheet, and Strip-Corrosion Resisting
TT-T-291	Thinner, Paint, Volatile Spirits (Petroleum-Spirits)
MMM-A-250	Adhesive, Water-Resistant, (for Sealing Fiberboard Boxes)
PPP-B-636	Box, Fiberboard
PPP-C-96	Can, Metal, 28 Gage and Lighter

Military

MIL-S-7952	Steel, Sheet and Strip, Uncoated, Carbon (1020 and 1025) (Aircraft Quality)
MIL-P-7962	Primer Coating, Cellulose-Nitrate Modified Alkyd Type, Corrosion-Inhibiting, Fast-Drying (for Spray Application over Pretreatment Coating)
MIL-C-8514	Coating Compound, Metal Pretreatment, Resin-Acid
MIL-A-18001	Anodes, Corrosion Preventive, Zinc; Slab, Disc and Rod Shaped
MIL-L-19537	Lacquer; Acrylic-Nitrocellulose, Gloss (for Aircraft Use)
MIL-S-22805	Spray Kit, Self Pressurized

STANDARDS

Federal

FED-STD-791	Lubricants, Liquid Fuels, and Related Products, Methods of Testing

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STANDARDS (continued)

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290	Packaging, Packing, and Marking of Petroleum and Related Products

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49 CFR 171-190	Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Qualification - The compounds furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Materials - The compounds shall be nonvolatile base materials dispersed in a solvent to form a fluid formulation conforming to this specification. Aromatic or chlorinated solvents shall not be used in the formulation. The compounds shall be homogeneous, free from grit, abrasives, water, chlorides, and other impurities.

3.2.1 Composition - The composition of the compounds shall be optional with the manufacturer but shall be within the limitations specified herein.

3.2.2 Application - Type I Class 1 and Type II Class 1 shall be capable of being applied by brushing, dipping, or spray gun, Type I Class 2 and Type II Class 2 shall be capable of being applied by pressurized container utilizing a fluorocarbon propellant conforming to BB-F-1421.

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3.2.3 Appearance - The applied compound shall form a film which is uniform and light brown or lighter in color. The film shall not exhibit froth, bubbling or excessive runoff.

3.2.4 Toxicity - When used for its intended purpose the material shall have no adverse effect on the health of personnel (see 6.5).

3.3 Solvent content - The solvent content of the compound shall consist of a non-photochemically reactive solvent blend. A non-photochemically reactive solvent is any solvent with an aggregate of less than 20 percent of its total volume composed of the chemical compounds classified below or which does not exceed any of the following individual percentage composition limitations, referred to the total volume of solvent:

a. A combination of hydrocarbons, alcohols, aldehydes, esters, ethers or ketones having an olefinic or cycloolefinic type of unsaturation: 5 percent;

b. A combination of aromatic compounds with eight or more carbon atoms to the molecule except ethylbenzene: 8 percent;

c. A combination of ethylbenzene, ketones having branched hydrocarbon structures, trichloroethylene or toluene: 20 percent

3.3.1 Toxicological data - The supplier shall furnish the toxicological data and formulations required to evaluate the safety of the material for the proposed use.

3.4 Properties of Type I Class 1 and Type II Class 1 - The corrosion preventive compound when tested as specified in Section 4, shall conform to the requirements of Table I.

3.5 Properties of Type I, Class 2 and Type II, Class 2 - The corrosion preventive compound, exclusive of propellants, when tested as specified in Section 4, shall conform to the applicable requirements of Table I and the following additional requirements when in pressurized cans with propellant.

3.5.1 Leakage - The pressurized cans shall not leak or become distorted when tested as specified in Section 4.

3.5.2 Fill - Pressurized cans containing the compound shall have a net weight of 16 ±0.4 ounces, when tested for fill as specified in Section 4.

3.5.3 Performance of pressurized containers - Compound packaged in pressurized containers shall spray uniformly, adhere to the panel, and shall not foam excessively or "sag" when tested as specified in Section 4.

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TABLE I

Properties of Types I and II Bulk Compound

	REQUIREMENTS		
Property	Type I Bulk	Type II Bulk	Test Paragraph
Flash point	140°F(min)		4.8.1
Dryness	0.0100g.(max)	-----	4.8.2
Synthetic sea water sulfuric acid spray	No visible corrosion of carbon steel after 2 cycles		4.8.3
Synthetic sea water displacement	No visible corrosion		4.8.4
Removability	Not more than 3 cycles to remove		4.8.5
Abrasives	None present		4.8.6
Film thickness	0.0005 inches(max)		4.8.7
Sprayability	Sprayable		4.8.8
Corrosivity	No visible pitting, etching, or dark discoloration. No weight change (milligrams/cm^2) greater than 0.5 for magnesium, cadmium and zinc nor greater than 0.2 for aluminum, copper and brass.		4.8.9
Staining	No visible evidence of staining or other deleterious effects		4.8.10
Dielectric breakdown	25000 volts(min)		4.8.11
Mixability	-------	No evidence of separation	4.8.12
Lubricity		Less than 0.20	4.8.13

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3.6 Workmanship - The workmanship shall be in accordance with high grade commercial practice covering this type of material. The ingredients shall be uniformly processed to produce the quality of products established by this specification. The finished products shall be clean and uniform, and free of any defects which might impair their utility such as sediment, metal, or other foreign particles in containers, eduction tubes and valve assemblies. When furnished in pressurized containers, the exterior orifice shall be symmetrical and free of ragged edges, and the exterior orifice, if drilled, shall be symmetrical and in direct alignment with angle of discharge.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections - The inspection requirements specified herein are classified as follows:

(a) Qualification inspection - Qualification inspection shall consist of those examinations and tests performed on samples submitted for approval as qualified products.

(b) Quality conformance inspection - Quality conformance inspection shall consist of those examinations and tests performed by the manufacturer on samples of compound which he intends to offer the Government under a contract or order.

4.2.1 Qualification inspection samples - Qualification inspection samples shall consist of the following:

Types I and II Bulk Form (Class 1) - Two gallons of the compound.

Types I and II Pressurized Container (Class 2) - Two gallons of the compound exclusive of propellants and five 16-ounce gas pressurized cans of the compound.

Samples shall be forwarded to the Supply Officer, Naval Air Development Center, Johnsville, Warminster, Pennsylvania 18974, marked, Attention: Director, Aero Materials Laboratory (Code 30223). Samples shall be plainly identified by securely attached durable tags marked with the following information:

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CORROSION PREVENTIVE COMPOUND, WATER
DISPLACING, ULTRA-THIN FILM
Type and Class Samples for Qualification Tests
Name of Manufacturer (Plant in which material is manufactured)
Manufacturer's Designation
Date of Manufacture
Submitted by (Name) (Date) for Qualification tests in accordance with the requirements of MIL-C-0081309C(AS) under authorization (reference authorizing letter)

The manufacturer shall submit a copy of test results with the qualification samples showing conformance with all the requirements of this specification and the applicable requirements of the Department of Transportation.

4.3 Qualification inspection - Qualification inspection shall consist of all the examinations and tests of this specification.

4.3.1 Retention of qualification - The retention of qualification of products approved for listing on the Qualified Products List (QPL) shall be maintained by periodic verification to determine compliance of the qualified product with the requirements of this specification. Unless otherwise specified by the activity responsible for the Qualified Products List, periodic verification shall be by certification and such certification shall be at intervals of not more than two years.

4.4 Quality conformance inspection - Qualification conformance inspection shall consist of examinations listed in 4.5.2, and the following tests:

Flash point
Dryness (for Type I)
Synthetic sea water - Sulfurous acid spray
Synthetic sea water displacement
Sprayability
Lubricity (for Type II)
Fill (for Class 2)

4.5 Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated herein.

4.5.1 Inspection lot -

4.5.1.1 Type I, Class 1, and Type II Class 1 - When the procurement document specifies delivery in bulk (see 6.2), an inspection lot shall consist of all compound produced from the same ingredients, prepared as a unit of manufacture, and offered for acceptance at one time.

4.5.1.2 Type I, Class 2, and Type II Class 2 - An inspection lot shall consist of all the pressurized containers, filled under basically the same conditions with compound produced from the same ingredients, offered for acceptance at one time.

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4.5.2 Examination of product -

4.5.2.1 Sampling for visual inspection - A random sample of filled containers shall be selected from each inspection lot in accordance with MIL-STD-105 at Inspection Level I and an AQL of 2.5 percent defective to verify conformance to all requirements of this specification regarding material (3.2).

4.5.2.2 Inspection of preparation for delivery - The packaging, packing, and marking of the compound shall be inspected to determine conformance to the requirements of Section 5 of this specification. Inspection Level shall be S-2 and the AQL shall be 2.5 percent defects per 100 units. Sample unit for this examination shall be a full container. Containers used for this examination shall be the same as those selected for visual inspection, and samples for tests.

4.5.2.3 Samples for test - Samples shall be selected in accordance with MIL-STD-105, Inspection Level S-3 with an AQL of 4.0 percent defective. Sample unit for test shall be one gallon of Type I,Class 1, and Type II Class 1. For Type I, Class 2 and Type II Class 2, the sample unit shall be one 16 ounce gas pressurized container and one gallon of the compound from which these samples were filled, exclusive of the propellant.

4.6 Test conditions - In general, physical tests contained in this specification shall be made under controlled atmospheric conditions having a relative humidity of 50 $\pm$5 percent and a temperature range of from 70° to 80°F. Waiver of this requirement may be permitted where proper conditioning facilities are not available for control testing. However, for referee purposes, the specified tests shall be made upon the compound under the specified atmospheric conditions.

4.7 Test disks and panels - (Except for Corrosivity and Staining)

4.7.1 Materials - The material for the test disks and panels shall be carbon steel conforming to FS1020 of MIL-S-7952.

4.7.2 Size of test disks and panels - Test panels, for tests requiring compound coatings, shall be 2 by 4 by 1/8 inches; except for the Synthetic sea water - Sulfurous acid spray and dryness tests where disks with a diameter of 2-1/8 inches and a thickness of 1/16 inch shall be utilized.

4.7.3 Preparation of test panels (or disks) - Panels (or disks) shall have all sharp edges and burrs removed and shall have all holes chamfered to prevent injury in handling. They shall be surface ground and hand polished with a 240 grit silicon carbide or aluminum oxide cloth or paper to produce a surface finish of 10 to 20 microinches (rms). Iron oxide or so-called "wet or dry" papers or cloths shall not be used.

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4.7.4 Cleaning of test panels (or disks) - The utensils and cloths used in the cleaning of test panels (or disks) shall be clean and free of contamination. Solvents shall be fresh and renewed frequently. In all stages of treatment the handling of panels (or disks) with the bare hands shall be avoided. The panels (or disks) shall not be permitted to contact contaminated surfaces during the cleaning procedure and shall be handled by tongs and hooks during and after dipping. After polishing, they shall be cleaned with a surgical gauze swab, in a beaker of hot mineral spirits conforming to Grade I of TT-T-291. Cleaning and scrubbing shall be followed by dipping in (1) a second container of hot mineral spirits, (2) boiling 95 percent methanol, and (3) boiling absolute methanol. The panels (or disks) shall be allowed to dry and shall then be stored in a desiccator until ready for use. If storage of more than 24 hours occurs, the surface preparation shall be repeated starting with the hand polishing.

4.7.5 Coating of the test panels (or disks) - Application of the compound to the test panels (or disks) shall be carried out under the atmospheric conditions of 4.6. A well-mixed representative sample of the compound shall be poured into a clean dipping chamber. The dipping chamber shall be of sufficient size to permit complete immersion of the panels (or disks). Care shall be taken to eliminate the formation of entrapped air bubbles by skimming them from the top of the compound before immersing the panels (or disks). They shall be coated by immersing them vertically in the compound and they shall remain completely immersed in the compound for one minute. Then they shall be removed singly at a rate of 4 inches per minute. After removal they shall be conditioned for 24 hours under the atmospheric conditions of 4.6 in a draft-, dust-, and fume-free atmosphere.

4.8 Test methods -

4.8.1 Flash point - Determination of flash point shall be conducted in accordance with Method 1103.6 of FED-STD-791.

4.8.2 Dryness (Type I) - Three test disks prepared as specified in 4.7.3 shall be cleaned as specified in 4.7.4, coated as in 4.7.5, using compound conditioned as in 4.8.3.1.2 and allowed to hang in a vertical position for one hour. They shall then be weighed and completely immersed vertically in talcum powder and withdrawn immediately. They shall then be reweighed to the nearest 0.0001 gram. The average change in weight shall be recorded. This procedure shall be repeated with test panels which have not been coated. These shall be used as controls. The average weight increase of the coated panels as compared with the weight increase of the uncoated panels shall be the measure of dryness.

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4.8.3 Synthetic sea water-sulfurous acid spray test -

4.8.3.1 Materials -

4.8.3.1.1 Test solution - The spray test solution shall consist of a solution made by adding 2 milliliters of sulfurous acid (6.4% assay as SO_2) to each liter of synthetic sea water. The pH shall be measured and shall be between 3.3 and 3.5. If it does not, additional acid or synthetic sea water must be added to adjust the pH to this range. The synthetic sea water shall be prepared by adding 50 grams of sodium chloride (NaCl); 22 grams of magnesium chloride ($MgCl_2.6H_2O$), 3.2 grams of calcium chloride ($CaCl_2.2H_2O$) and 8.0 grams of sodium sulfate (Na_2SO_4) to a liter of distilled or demineralized water. One liter of fresh test solution shall be used for each cycle.

4.8.3.1.2 Conditioning of compound sample - A wide mouth, quart jar containing 800 ml of the compound shall be sealed and exposed to the following cycle four times without physical disturbance of the compound:

Eight hours at 130 ±2°F
Sixteen hours at -40 ±2°F

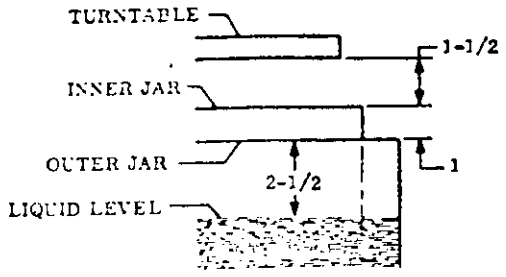
At the end of this cyclic exposure the contents of the sealed jar shall be allowed to remain at 75 ±5°F for 24 hours.

4.8.3.2 Type I -

4.8.3.2.1 Apparatus - The apparatus used in this test should conform to Figures 1, 2, 3, 4, and 5. The apparatus may be combined in one cabinet and made fully automatic or may be manually operated as two separate units.

4.8.3.2.2 Procedure - Test disks prepared as specified in 4.7.3 shall be cleaned as specified in 4.7.4, coated as specified in 4.7.5, using compound conditioned as in 4.8.3.1.2 allowed to dry overnight in a horizontal position on a level surface and shall then be placed in the disk holders. The temperature of the water jacket shall be 100 ±2°F when the test is started and shall be maintained at this temperature during the one hour spray. The disk holders shall be placed in the turntable and the turntable started. The air, regulated to a flow of one ±0.1 cubic foot per minute shall be supplied to the nozzle to effect the spray. The spray shall continue for one hour at which time the spray is shut off and the disk holders containing the test disks shall be moved to a position over the 275 watt ultra violet sunlamp. The lamp shall be then turned on for three hours. The combination of one hour spray and three hours exposure to the ultra violet lamp shall be defined as one cycle. At the completion of the second cycle the disks shall be cleaned in mineral spirits conforming to Grade 1 of TT-T-291 after which they shall be examined for any visible corrosion, pitting or staining under 10X magnification.

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DETAIL "A"

LEGEND

1. Acrylic Turntable, positioned with the lower surface 1-1/2 inches above the inner jar rim (4) and 2-1/2 inches above the outer jar rim (3). The turntable (see Figure 2) is provided with holes for the mounting of the test disk holders. The turntable is mechanically rotated, driven by the electric motor (2) encased in a protective acrylic box, mounted above the outer and inner jars.
2. Electric Drive Motor
3. Outer Jar, 16 inches Diameter, 12 inches High
4. Inner Jar, 12 inches Diameter, 12 inches High
5. Liquid Fill Tube
6. Plastic Tubing
7. Air Inlet From Vibrator Pump
8. Air Disperser
9. Liquid Pickup Tube
10. Drain Tube Assembly
11. No. 13 Rubber Stopper
12. Thermo Regulator
13. Immersion Heater (250 W)
14. 110 Volt Heater Cord
15. Compressed Air Inlet, 12 psi, 1 ft.^{3}/min.
16. Acrylic Atomizer Support - See Figure 3
17. Atomizer Assembly - See Figure 4

NOTE: Inner Jar Must Be Weighted To Prevent Floating
DIMENSIONS ARE IN INCHES

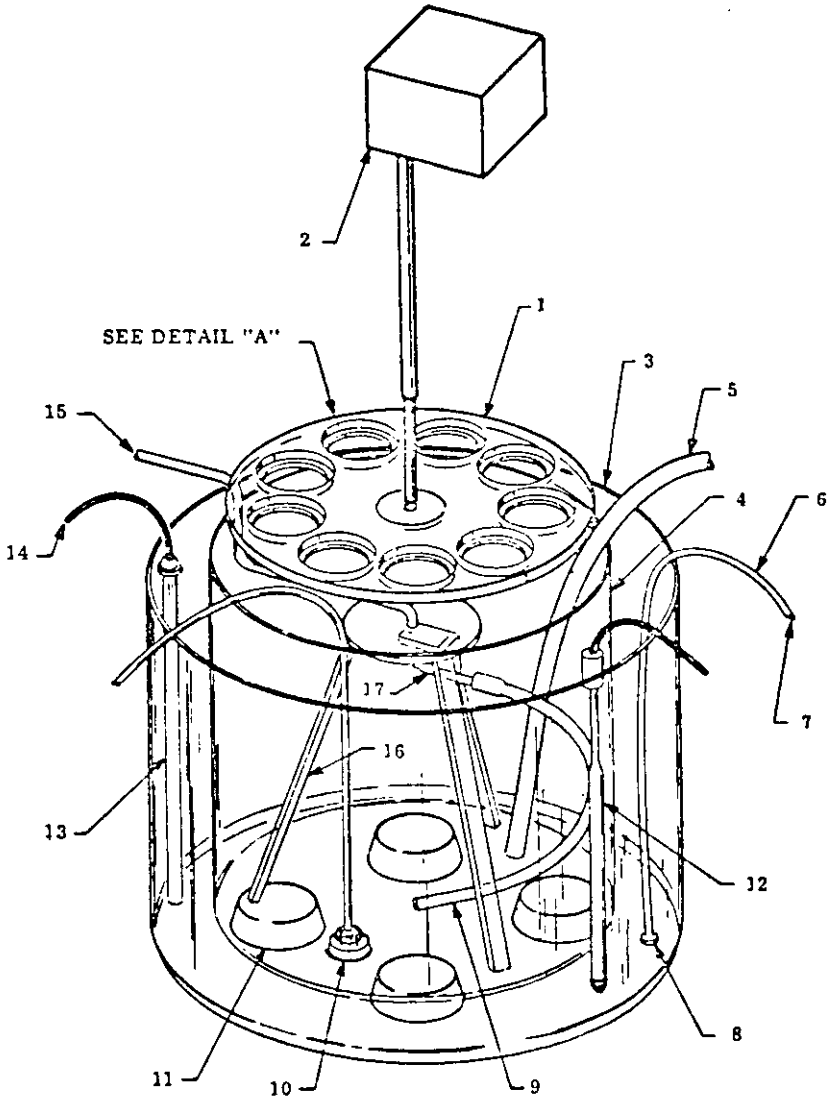
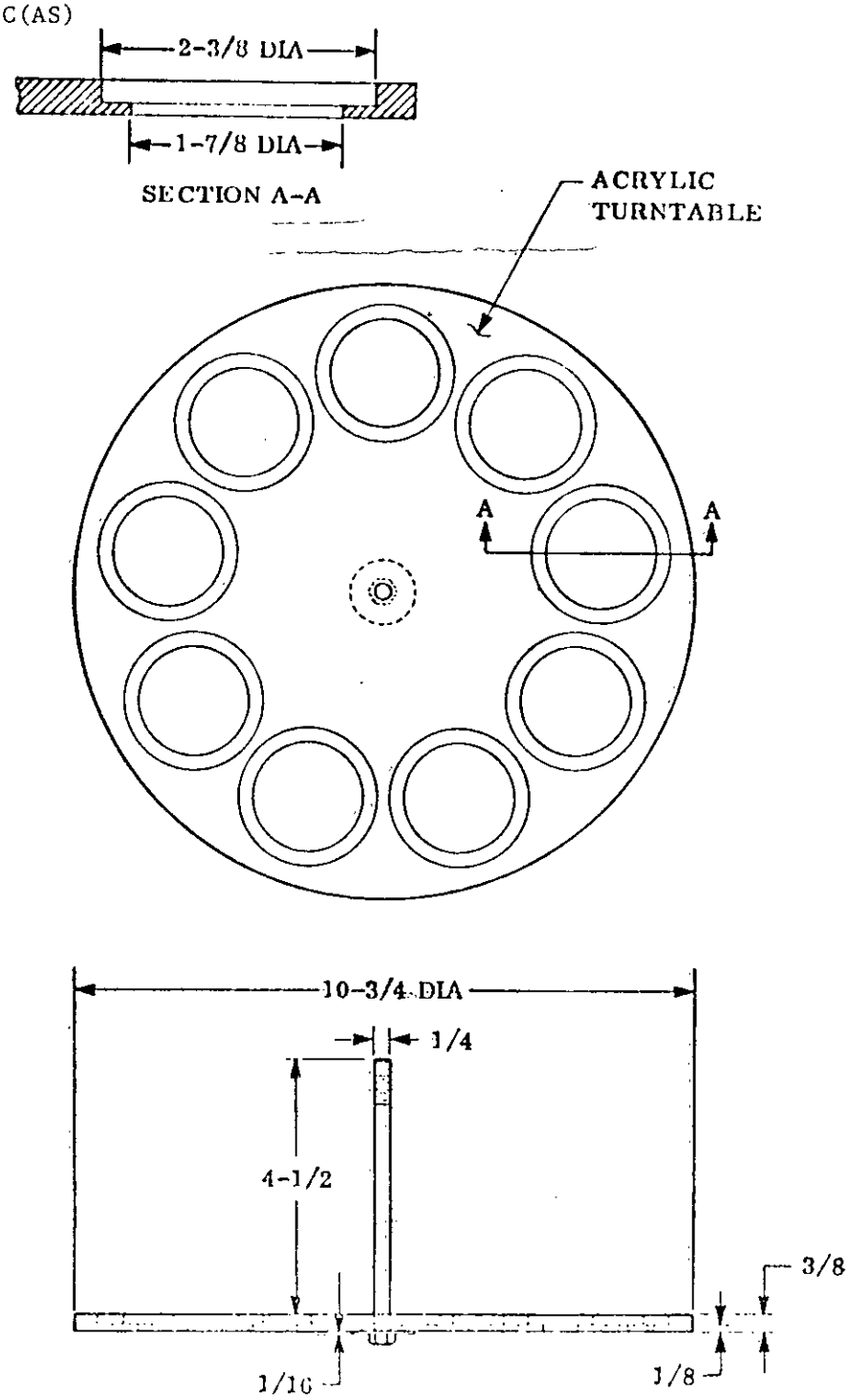


Figure 1. Synthetic Sea Water - Sulfurous Acid Spray Test Apparatus

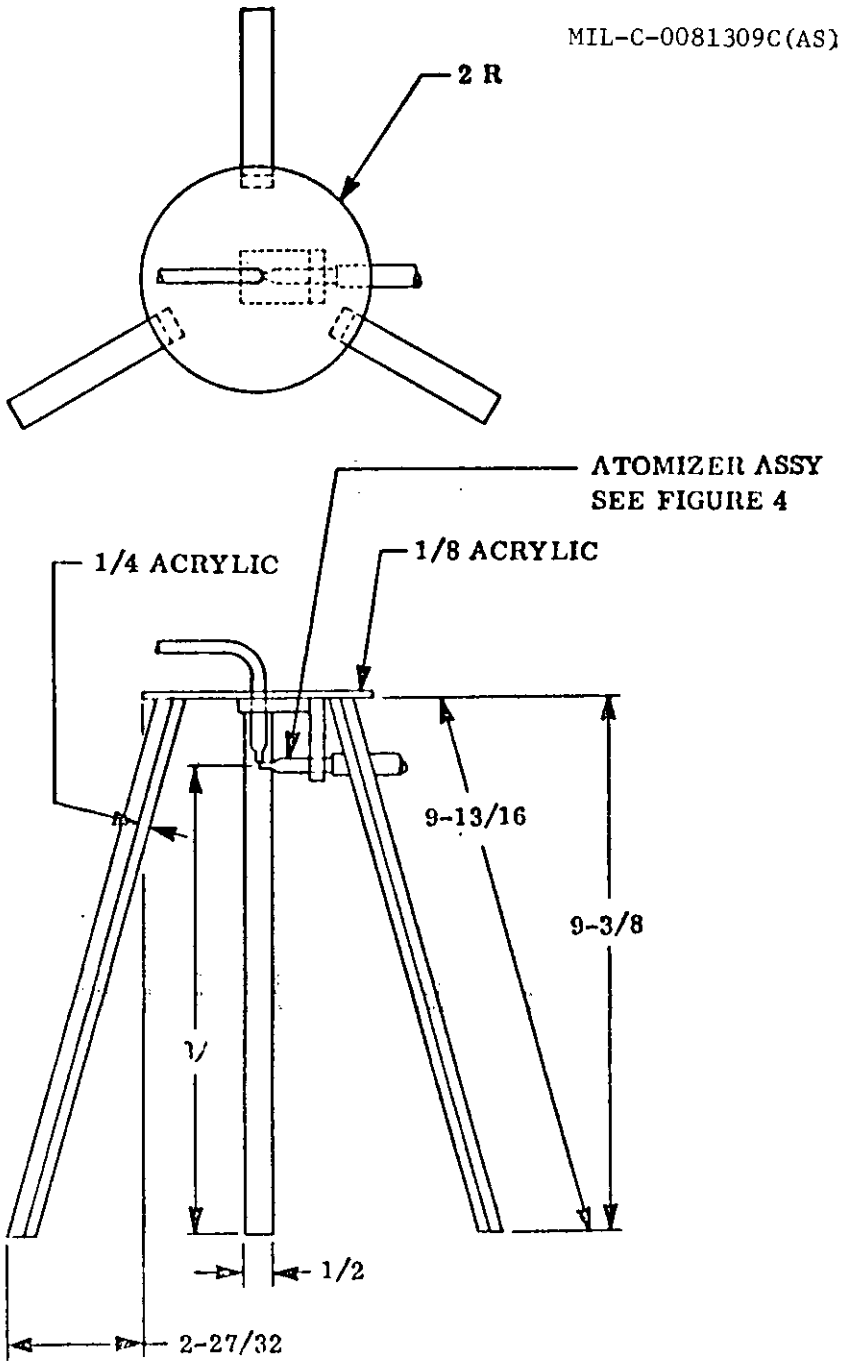
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DIMENSIONS ARE IN INCHES

Figure 2. Acrylic Turntable Detail

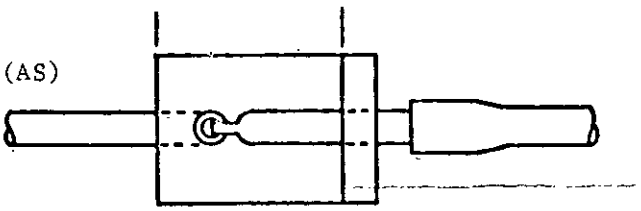
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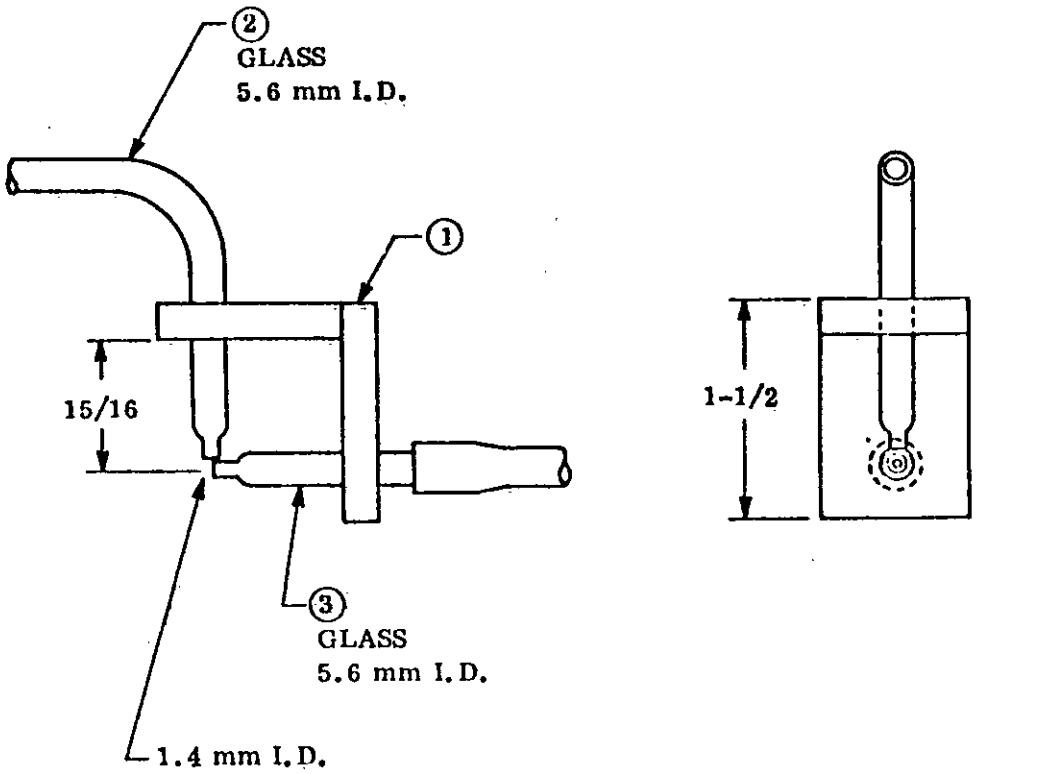


DIMENSIONS ARE IN INCHES

Figure 3. Atomizer Support Assembly

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① 1/4" ACRYLIC MATERIAL
② FILTERED AIR FLOW
③ LIQUID PICK-UP-TUBE

DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED

Figure 4. Atomizer Assembly

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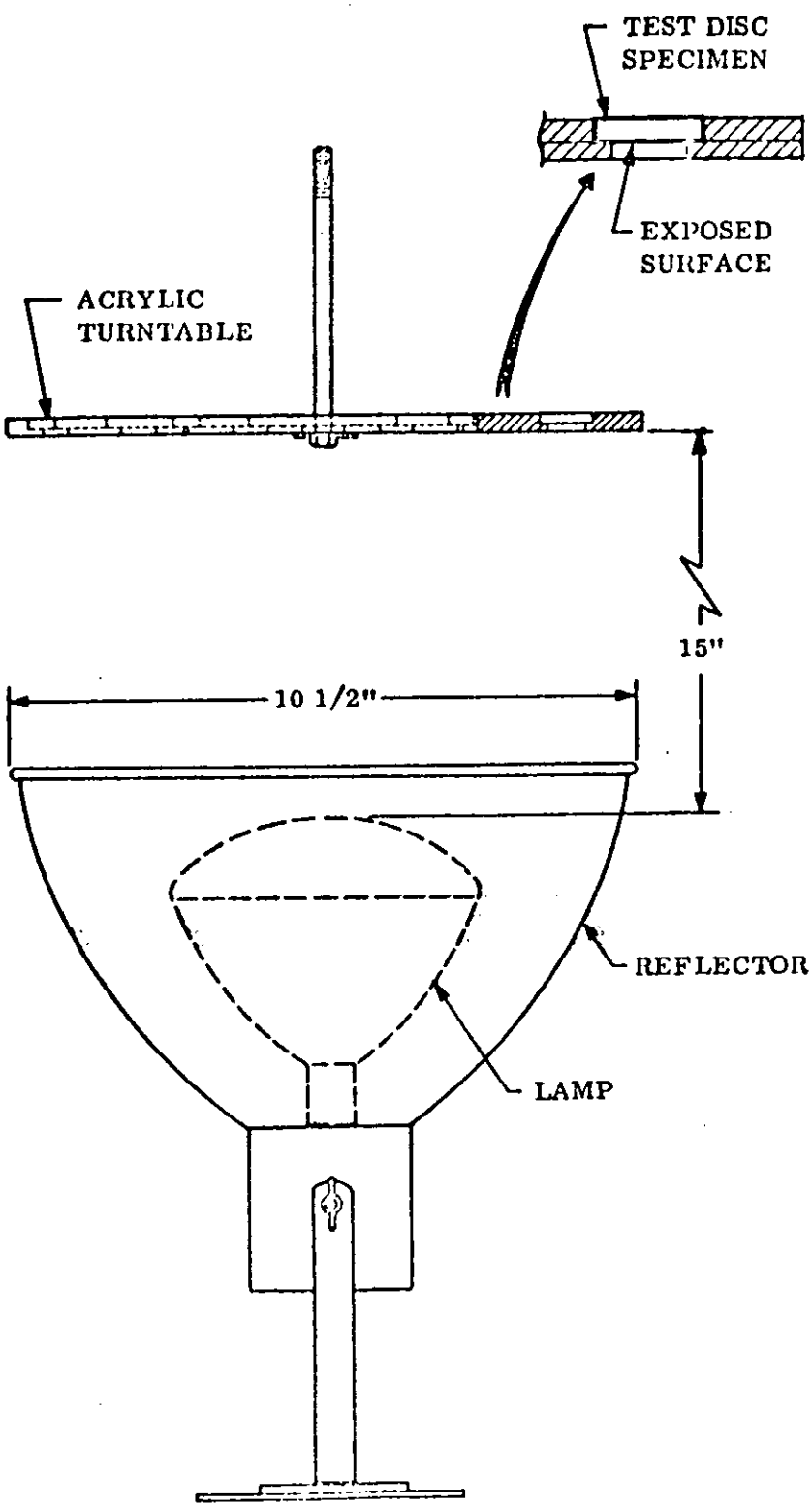


Figure 5. Ultraviolet Sunlamp Exposure

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4.8.3.3 Type II Compound -

4.8.3.3.1 Apparatus - The apparatus used in the test shall conform to Figures 6, 7, and 8.

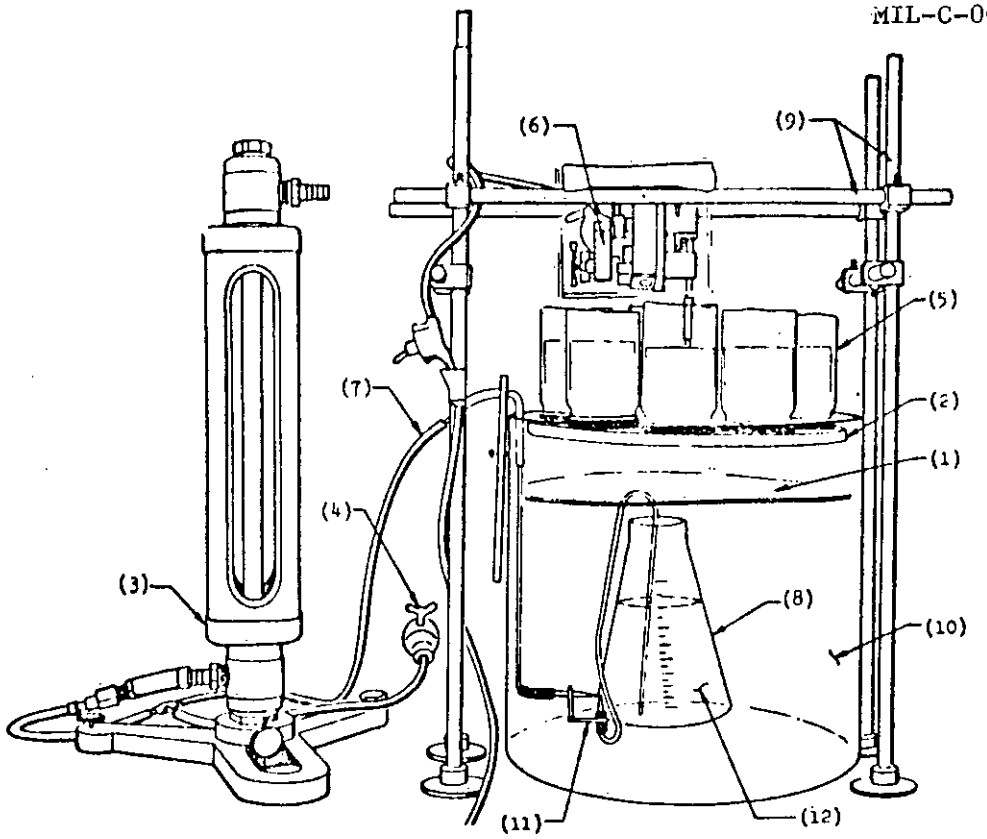
4.8.3.3.2 Procedure - Applicable test disks prepared as specified in 4.7.3 shall be cleaned as specified in 4.7.4, coated as specified in 4.7.5, using compound conditioned as in 4.8.3.1.2 allowed to remain overnight in a horizontal position on a level surface and shall then be placed in the disk holders. The turntable shall then be started and the air, regulated to a flow of one ±0.1 cubic foot per minute, shall be supplied to the nozzle to effect the spray. The pickup tube supplying the nozzle should have an I.D. of 1 mm. At the end of approximately 2 minutes of spraying, 35 milliliters of water which has been cooled to 35°F shall be added to each water jacket to cause a fine condensate to form on the surfaces of the disks. Spraying shall continue for one hour. The rate of flow shall be approximately 1400 ml/hr. and during this 1 hour spray period, the sea water shall be replaced at the midpoint of the period. At the end of the hour the spray shall be stopped, the water jackets shall be removed, and the turntable shall be removed from the jar and placed where the disks may dry at room temperature for 3 hours. The combination of 1 hour spray and 3 hours drying shall be defined as one cycle. At the end of each cycle the jar shall be emptied of the solution. Two cycles shall be conducted in one day. The disks shall then be cleaned in mineral spirits conforming to Grade I of TT-T-291 after which they shall be examined for any visible corrosion, pitting, or staining.

4.8.4 Synthetic sea water displacement -

4.8.4.1 Procedure - Panels prepared as in 4.7.3 and 4.7.4 shall be placed so that one 2-inch end shall be raised one inch above a horizontal surface. The panels shall then be sprayed with the synthetic sea water of 4.8.3.1.1 so that the entire upper surface of the specimen is covered with tiny droplets. Within one minute after spraying, one milliliter of the test compound shall be poured along the upper two-inch edge of the panels and allowed to run slowly down the specimen so as to completely cover the test panel. After another minute, a second milliliter of the test compound shall be poured and allowed to run down the panels in a like manner. After waiting an additional minute, the panels shall be picked up and held in a vertical position for one minute and shall then be placed flat (test side up) above distilled water at 72°F in a closed desiccator. After 4 hours they shall be removed and cleaned with mineral spirits, and then evaluated for presence of visible corrosion.

4.8.5 Removability - The apparatus for determining removability shall be similar to Figure 9. Removability shall be determined on three test panels that have been prepared and cleaned as specified in 4.7.3 and 4.7.4 and coated as in 4.7.5. They shall be placed one at a time in a

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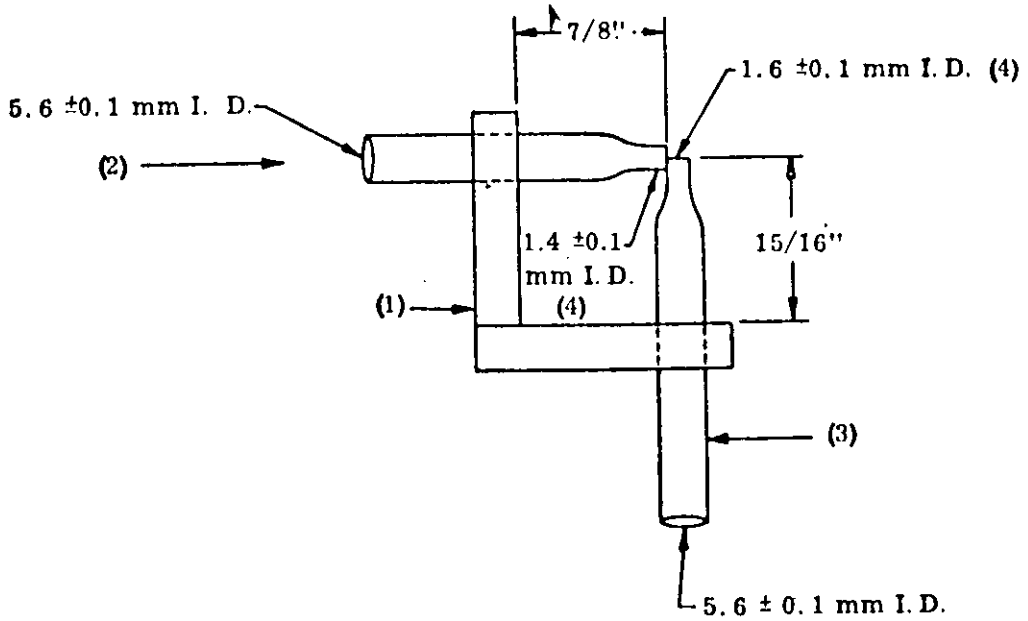


NOTES:

1. Acrylic Baffle - 12 inches in diameter or snug fit in jar (10), with a 3-inch diameter center hole. The baffle is located within the jar (10) above the spray nozzle (11), thus preventing direct impingement of the spray on the disks, but allowing the fine spray mist full contact with the disks under test. Edge of baffle shall be notched to allow passage of the tube leading to spray nozzle (11).
2. Acrylic Turntable - 10-1/2 inches in diameter, positioned approximately 1/2 inch below the top of the jar (10). The turntable is provided with holes for the mounting of the test disk holders (5) and disks. The turntable is mechanically rotated, driven by an electric motor (6) (encased within a protective acrylic box, mounted above the jar (10) on an arrangement of Flex-frame support rods (9)).
3. Air Regulator
4. Air Source
5. Coated Test Disk Holder - See Figure 10
6. Electric Motor (slow speed one r.p.m. or less)
7. Filtered Air Source - With an air regulator (3) capable of regulating the air flow to the spray nozzle (11) at one cubic foot per minute
8. Flask - 1000 ml, extra wide mouth, Erlenmyer flask, placed in the jar (10) to hold the synthetic sea water-sulfurous acid solution of 4.8.2.2.1 (see Note 12)
9. Flex-Frame Support Rods
10. Jar - 12 inches I.D. diameter and 12 inches high
11. Spray Nozzle - See Figure 2
12. Test Solution - See 4.8.2.2.1

Figure 6. Synthetic Sea Water - Sulfurous Acid Spray Test Apparatus

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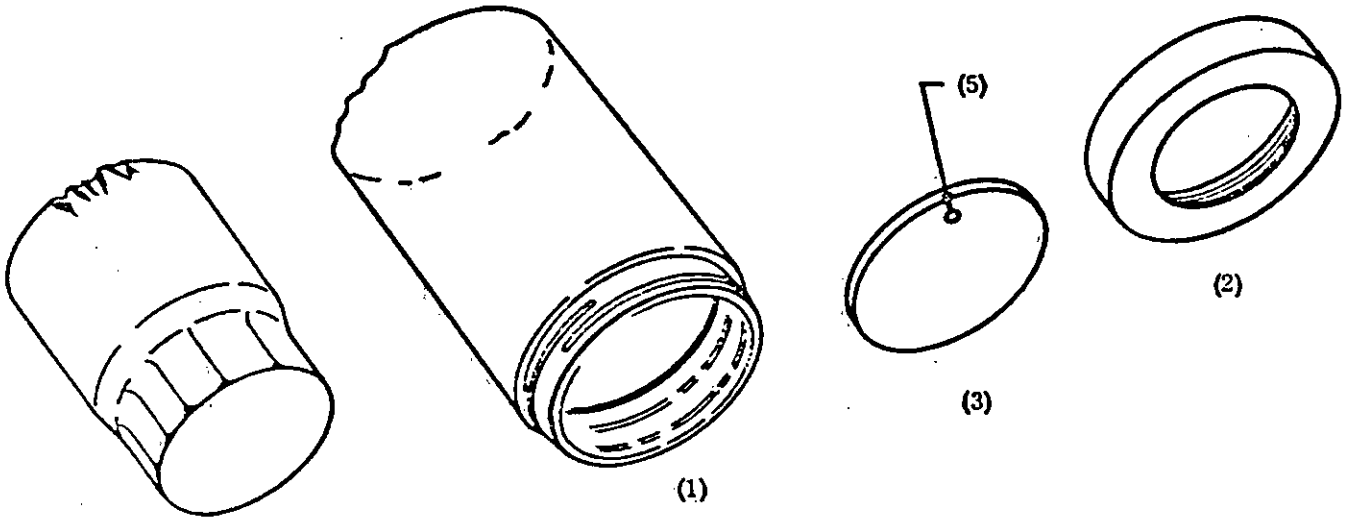
NOTES:

1. 1/4" Acrylic Material.
2. Filtered Air Source
3. Liquid Pick-Up Tube
4. Orifice

General - The Spray Nozzle is provided with suitable piping for spraying from the flask (see Note 8, Figure 1). The nozzle is powered by a filtered air source (2). (See Note 7, Figure 1)

Figure 7. Spray Nozzle

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NOTES:

1. Disk Holder
2. Disk Holder Cap
3. Test Disk - A 1/32 inch diameter hole (5) is provided for handling the disk by stainless steel wire.
4. Water Jacket - In use, the aluminum water jacket is placed so as to be in direct contact with the back of the round test disk of 4.7.2. ((3) above.)

General - The complete test disk holder, with disk, described above, is mounted in one of the holes of the acrylic turntable of the apparatus in Figure 1 (see Notes 2 and 5, Figure 1).

Figure 8. Water Jacket; Test Disk, Holder and Cap

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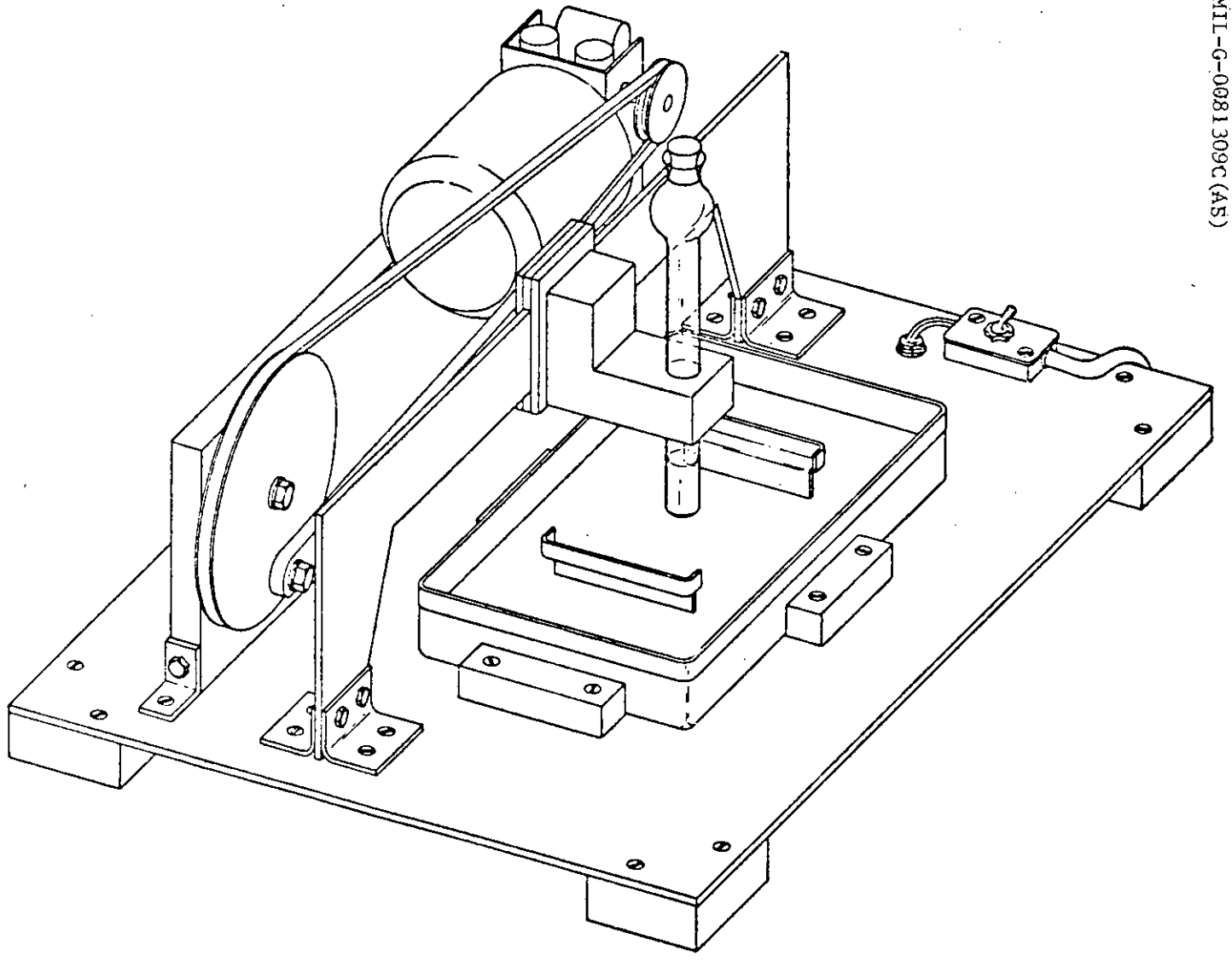


Figure 9. Apparatus for Determining Removability

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position in the apparatus so that a felt pad wiper, saturated with solvent conforming to Grade I of TT-T-291, can wipe a streak 3 inches long, in a forward and backward motion, on the center of the panels. The wiper shall consist of a 2-inch long wick made from an all wool felt (weighing approximately 190 ounces per square yard in a 3/4 inch thickness) which shall be packed tightly into the lower end of a tube (wiper holder) 10 inches long, and 5/16 inch inside diameter and shall protrude 1/4 to 1/2 inch. One or more indentations shall be made approximately 2 inches from the bottom of the wiper holder tube to prevent the wiper from being forced further into the tube. The wiper holder shall fit into a supporting tube 4 inches long and 7/16 inch inside diameter which is fixed in position and acts as a sleeve or guide. The wiper holder tube shall be filled with solvent and tightly stoppered. It shall be opened and vented before each run so that the wiper felt is filled with solvent in an amount sufficient to make the run but without allowing the solvent to flow freely over the panel. The length of the wiping stroke shall be 3 inches and the speed of the machine shall be regulated to 40 $\pm$5 cycles per minute. A complete wiping cycle shall consist of a stroke of approximately 3 inches in one direction with a return stroke to the starting position. At the completion of the third cycle, the compound under the wiper shall be completely removed (determined visually) over an area 1/4 inch wide and the entire length of the stroke.

4.8.6 Abrasives - Mix approximately 75 milliliters of the compound with 200 milliliters of benzol and stir until all soluble matter is in solution. Allow to stand for one hour at room temperature to permit any insoluble matter to settle. Carefully decant, wash the residue with 100 milliliters of benzol and again carefully decant. Repeat the procedure with successively smaller portions of benzol until the solution is practically colorless. The residue after the last decantation shall be rubbed between two pieces of flat clean glass plate. The appearance of scratches on the glass plate shall be considered evidence of the presence of abrasive material.

4.8.7 Film thickness - The average film thickness produced by the compound shall be determined on a panel prepared, cleaned, and coated as specified in 4.7.3, 4.7.4, and 4.7.5 respectively. The film thickness shall be calculated using the following formula:

$$\text{Film thickness (mils)} = \frac{(1{,}000 \times W) \times 0.061}{D \times A}$$

Where: W = weight of film (grams)
0.061 cubic inch = 1 cubic centimeter
D - density of the film (grams per cubic centimeter) Density is determined on the nonvolatile portion by means of a pycnometer
A - total surface area of the panel (square inches)

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4.8.8 Sprayability - Three ounces of the compound shall be placed in a 4 ounce product jar conforming to MIL-S-22805 and conditioned at room temperature for three hours. Spray the material using a spray device conforming to MIL-S-22805 that has been conditioned at room temperature. The material shall be considered as having passed the test if it can be satisfactorily sprayed.

4.8.9 Corrosivity -

4.8.9.1 Preparation of specimens - Specimens of the following metals shall be used in this test:

Magnesium, QQ-M-44
Cadmium, QQ-A-671
Zinc, MIL-A-18001
Aluminum, QQ-A-250/4
Copper, QQ-C-576
Brass, QQ-B-626

Suggested specimen size is 3 x 1/2 x 1/16 inches.

4.8.9.2 Immersion - Three specimens of each of the above metals shall be polished to remove pits, burrs, and irregularities from all faces and edges. The panels shall be finished and cleaned as specified in 4.7.3 and 4.7.4. After weighing, the specimens shall be placed in screw cap jars with one specimen in each jar. The specimens shall be placed on one of their narrow ends. The specimens shall be covered with enough compound so that the tops of the specimens are at least 1/4 inch below the surface of the compound. The sealed jar shall then be placed in an oven at 130 $\pm$2°F for 7 days. Upon completion of the test, any loose corrosion products shall be removed from the specimens by cleaning as specified in 4.7.4. Reweigh the specimens and calculate the weight loss or gain in milligrams per square centimeter.

4.8.10 Staining -

4.8.10.1 Preparation of test panel - A test panel measuring 3 by 6 by 0.020 inches shall be fabricated from aluminum alloy conforming to QQ-A-250/5. The panel surface shall be prepared as follows: Spray one coat of wash primer, conforming to MIL-C-8514, to a dry film thickness of 0.0002 to 0.0003 inch and air dry for 30 minutes. One coat of lacquer primer, conforming to MIL-P-7962, shall be sprayed over the wash primer to a dry film thickness of 0.0003 to 0.0004 inch and air dried for 30 minutes. Two coats of acrylic nitrocellulose insignia white lacquer, conforming to MIL-L-19537, shall be sprayed over the preceding films, allowing a 30-minute drying time between coats. The panel shall be permitted to dry at least 24 hours in a clean dust-free atmosphere before performance of test.

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4.8.10.2 Test procedure - Three smears of the compound shall be applied to the test surface. The smears shall run parallel to the 3-inch dimension of the panel and shall measure 3/4 to 1 inch wide by approximately 0.005 inch thick. They shall be so spaced as to permit ready comparison with the uncoated paint surface upon removal of the compound. The panel with the applied smears shall be placed under a 275 watt ultraviolet R.S. sunlamp. The center of the sunlamp shall be positioned 15 inches from the center of a 14 inch diameter turntable which shall revolve at an approximate rate of 3 rpm. The test shall be conducted at 75 ±5°F. A constant line voltage to the lamp shall be maintained so that a direct scale reading of 20 units is obtained on a General Electric sunlamp tester No. 113 or its equivalent, used without shield. After a 72-hour exposure period the panel shall be degreased with mineral spirits, TT-T-291, Grade I applied with a clean gauze swab and the panel surface examined for stain. If a minimum of one of the test areas exhibits stain, the test shall be repeated. In the retest evidence of stain in one or more areas shall be cause for rejection.

4.8.11 Dielectric breakdown voltage - The compound shall be subjected to and shall pass the dielectric breakdown voltage inspection in accordance with the procedure of ASTM D877.

4.8.12 Mixability (Type II only) - Combine 5 ml of compound and 95 ml of lubricating oil (MIL-L-17672) in a 100 ml cone-shaped centrifuge tube. Shake by hand until the mixture appears homogeneous. Place the tube in a beaker and heat in an explosion-proof oven at 170° ±2°F for 15 minutes. There shall be no evidence of separation or sediment after the 15 minute 170°F temperature inspection or after 24 hours at 77° ±5°F.

4.8.13 Lubricity (Type II only) -

4.8.13.1 Apparatus - The test apparatus (Figures 10&11) shall consist of: (1) a 5 x 16 x 3/8 inch 2024 T6 milled aluminum plate with a surface roughness greater than 9 and less than 15 microinches (rms) when determined in the direction parallel to the grain of the plate, and greater than 14 and less than 20 microinches (rms) when determined across the grain; (2) a freely rotating wheel affixed to the plate by an axle parallel to the width approximately one inch from an end; (3) a block holding 3 test balls motionless in the configuration shown (Figure 10) and weighted in such a manner that each ball exerts a force equivalent to 250 grams upon the surface of the aluminum plate: and (4) a pan or beaker attached by a flexible string over the pulley wheel to the three ball holder so that when taut the string is parallel to the plate surface and is counterbalanced by the block.

4.8.13.2 Materials - The test ball bearings shall be of 1/2 inch diameter AISI-C 52100 steel, Grade 25 EP (extra polish) quality steel balls of the Anti-Friction Bearing Manufacturer's Association.

4.8.13.3 Procedure - One surface of the aluminum plate shall be coated with Type II corrosion preventive compound to a thickness of not

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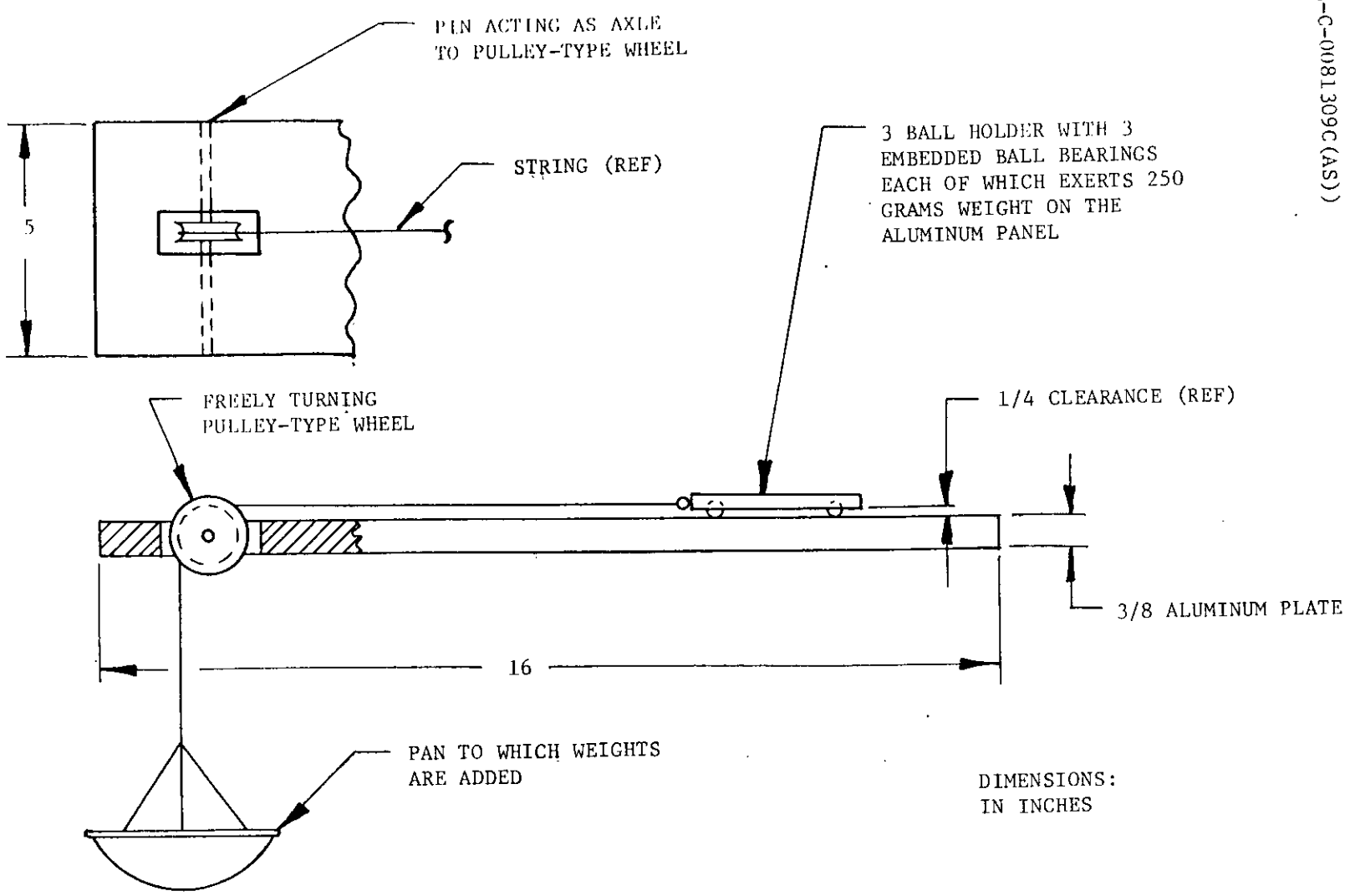
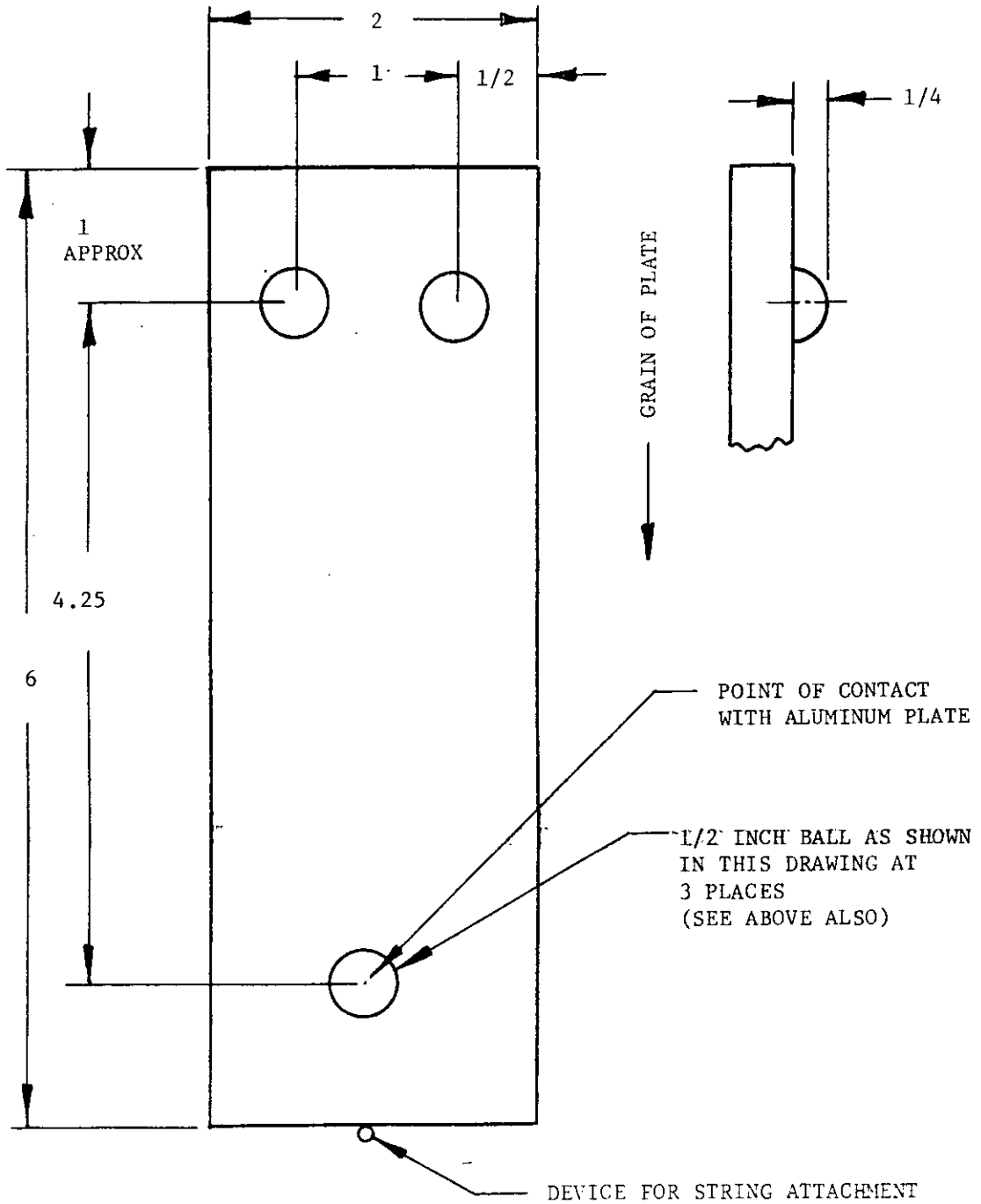


Figure 10

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DIMENSIONS:
IN INCHES
MATL: 1/2 THK
ALUMINUM

BOTTOM VIEW OF PLATE

Figure 11

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less than 0.001 inches and not more than 0.005 inches. The coating shall be exposed at 72°F for 24 hours. At the conclusion of the exposure period, the weighted three ball holder shall be set gently on the coated surface at the end of the plate opposite the pulley wheel with one ball forward and two rearward and oriented so that if pulled by the string attachment toward the pulley wheel no rotational motion would occur. The pan or beaker shall be attached as shown (Figure 10). Test balls shall be lightly dropped onto the pan from a height of less than 1/4 inch until the three ball holder moves. If the holder traverses a distance of six (6) inches in less than three (3) seconds, the gross weight of the pan shall be recorded. If the movement requires more than three seconds, the holder shall be placed as before at one end of the aluminum plate in a location different from those previous. Another test ball shall be dropped onto the pan from a height of less than 1/4 inch and the travel time observed. The restarting procedure shall be repeated until the three ball holder travels six inches in less than three seconds at which time the gross weight of the pan shall be recorded.

4.8.13.4 Calculation - The lubricity value shall be determined from the following formula:

$$\frac{\text{weight required}}{750\text{g}} = \text{lubricity value (coefficient of friction)}$$

4.8.14 Specialized tests for Class 2 - (Pressurized Containers)

4.8.14.1 Leakage - The pressurized container shall be completely submerged for 5 minutes in water maintained at a minimum of 130 ±2°F during which it shall be observed for the emission of bubbles. Distortion of the container or the emission of bubbles from any part of the container shall be considered evidence of leakage.

4.8.14.2 Fill - A sample can shall be weighed and then shall be sprayed at three minute periods with one-minute intervals until the can is exhausted. The container shall be re-weighed. The net difference shall be at least 16 ounces.

4.8.14.3 Performance of pressurized containers - Panels as described in 4.7.2 shall be used. A panel shall be supported such that the longer dimension forms a 45° angle with the horizontal. Corrosion preventive compound packaged in accordance with 5.2 shall be sprayed on the panel from a distance of 12 inches. The panel shall be examined for uniformity of spray, foaming, and adherence to the substrate. After a 10 second pause the same panel shall be resprayed and examined for adhesion and sagging. After a 5 second pause the same panel shall be resprayed again and likewise examined.

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5. PREPARATION FOR DELIVERY

5.1 Packaging - Except as specified in 5.1.1, packaging shall be Level A or C in accordance with MIL-STD-290 as specified in the contract or order. Neither the container, nor any component thereof (closure, lining, etc.), shall interact with or alter the contents in any way so as to adversely affect their purity or quality. All containers shall be new and free from contaminants.

5.1.1 Packaging of compound in gas pressurized containers - the pressurized containers shall hold 16 ounces. The containers shall conform to Class 2, Type IX of PPP-C-96 with a valve opening diameter suitable for the specified valve.

5.2 Packing - Except as specified in 5.2.1, packing shall be Level A, B, or C in accordance with MIL-STD-290 as specified in the contract or order (see 6.2).

5.2.1 Packing of filled pressurized containers for shipment -

5.2.1.1 Level A - Twenty-four dispensers shall be packed in a fiberboard box conforming to PPP-B-636, Style F.O.L., compliance symbol V3s or V3c. The twenty-four dispensers shall be arranged: six in length, four in width and one in depth, and shall be separated by slotted partitions providing an individual cell for each dispenser. Partitions shall be B or C flute, double faced corrugated board. Box liners of the same material as the partitions shall be provided. The corrugations of the liners shall run vertically. Liners shall be cut so that on placement the ends abut in the middle of one side of the box. Box and all components shall be fabricated of material having not less than 200 pounds per square inch bursting strength.

5.2.1.1.1 Closure for level A - All flaps of the box shall be securely sealed with a water-resistant adhesive conforming to MMM-A-250. The adhesive shall be applied throughout the entire area of contact between the flaps.

5.2.1.2 Level B - Twenty-four dispensers shall be packed in a domestic type corrugated or solid fiberboard container, Style F.O.L. (less 1 inch) conforming to PPP-B-636. Arrangement shall be: six in length, four in width, and one in depth. Slotted partitions shall be employed to form an individual cell for each dispenser. Partitions shall be B or C flute, double faced corrugated board. Box liners of the same material as the partitions shall be provided. The corrugations of the liners shall run vertically. Liners shall be cut so that on placement the ends abut in the middle of one side of the box. Box and all components shall be fabricated of material having not less than 200 pounds per square inch bursting strength. All flaps shall be sealed with a good quality adhesive applied throughout the entire area of contact between flaps.

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5.2.1.2.1 Strapping - Strapping shall be in accordance with the appendix of PPP-B-636.

5.3 Marking - Marking of the containers shall be in accordance with MIL-STD-290, except as specified herein. Marking shall be legible, shall be accomplished by lithographing or silkscreen process and shall be in white on a bright red label for Type I, Class 2; white on a bright green label for Type II, Class 2. Type I, Class 1, and Type II, Class 1 may be marked in black on a white background or as specified in the contract. Paper coated labels on pressurized containers are not acceptable. Any special marking specified in the contract or order shall also be included. In addition, the following information shall be included on each gas pressurized container (when not already required by MIL-STD-290 or contract or order):

Front Face (Type I, Class 2 only):
DRY-TO-THE-TOUCH
(Stock No.)
CORROSION PREVENTIVE COMPOUND,
WATER DISPLACING, ULTRA-THIN FILM
MIL-C-0081309C(AS)
(TYPE I, CLASS 2)
(CONTRACT NO.)
(MANUFACTURER'S NAME)
(MANUFACTURER'S ADDRESS)
(MANUFACTURER'S PRODUCT NO.)
16 OUNCES NT. WT.

Front Face (Type II, Class 2 only):
SOFT FILM
(STOCK NO.)
CORROSION PREVENTIVE COMPOUND
WATER DISPLACING, ULTRA-THIN FILM
MIL-C-0081309C(AS)
(TYPE II, CLASS 2)
(CONTRACT NO.)
(MANUFACTURER'S NAME)
(MANUFACTURER'S ADDRESS)
(MANUFACTURER'S PRODUCT NO.)
16 OUNCES NT. WT.

Rear Face:
FOR BEST RESULTS:
Hold can approximately 12 inches from the surface to be covered. Spray a thin, uniform coating.

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6. NOTES

6.1 Intended use - The ultra-thin film, corrosion preventive compounds covered by this specification is intended for use on any metal surfaces. It can be used both for initial protection during shipment and storage and for in-service treatment. It should not be used around liquid oxygen fittings. The ability of these materials to prevent corrosion and to displace water and their ease of application when packaged as Class 2 (pressurized spray cans) make them particularly suited for service use.

6.1.1 Type I, Classes 1 and 2 - This compound is intended for use on non-moving parts not requiring a lubricated surface, such as fasteners, seams, doors, joints, unpainted metal and where paint is cracked or damaged.

6.1.2 Type II, Classes 1 and 2 - This compound is intended for use on moving parts where some lubrication is required such as hinges, bomb racks, and sliding parts and also on certain avionics equipment such as electrical connector plugs.

6.2 Ordering data - Requests, requisitions, schedules, and contracts or orders should specify the following:

(a) Title, number, and date of this specification.

(b) Quantity. (For bulk material specify gallons; for pressurized containers specify number of 16-ounce cans.)

(c) Type and Class.

(d) Packaging desired (see 5.1).

(e) Level of packing required (see 5.2).

(f) Labeling or other special marking required (see 5.3).

6.3 Qualification - With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360; however, information pertaining to qualification

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of products may be obtained from the Director, Code 30223 Naval Air Development Center, Warminster, Pennsylvania 18974.

6.3.1 It is to be understood that upon receipt of the Letter of Authorization, samples shall be furnished at no cost to the Government, and that the manufacturer shall pay the transportation charges to and from the designated point where tests are to be made. In the case of failure of the sample or samples submitted, consideration will be given to the request of the manufacturer for additional tests only after it has been clearly shown that changes have been made in the product which the Government considers sufficient to warrant additional tests, and a new designation is given the material by the manufacturer.

6.4 It is understood that Section 173.306 of the Interstate Commerce Commission Regulations specifies that each completed metal container filled for shipment must be heated until the contents reaches a minimum temperature of 130°F without evidence of leakage, distortion, or other defects.

6.5 Any questions raised regarding toxicity should be referred by the procuring agency to the departmental medical authority. In Army procurement, the Surgeon General will act as advisor to the procuring agency.

6.6 Relationship of this specification to previous issues - MIL-C-81309A provided two grades of soft film corrosion preventive compound; Grade A for use on all metals and Grade B for use on stainless steel and aluminum in interior locations. MIL-C-0081309B(AS) provides one grade of dry-to-touch compound which can be used on all metals. Under both of these specifications, bulk material or pressurized spray (aerosol) cans were procured on order but were not classified separately. MIL-C-0081309C(AS) provides for both dry-to-touch films and soft film compounds for applications as described in 6.1 above. In addition, provision was made to facilitate procurement operations by defining separate classes for bulk or pressurized spray cans (aerosols). The equivalents are as shown below:

Table of Equivalents

MIL-C-0081309C(AS)	Previous Specifications
MIL-C-0081309C(AS) Type I, Class 1	MIL-C-0081309B(AS) (Bulk)
MIL-C-0081309C(AS) Type I, Class 2	MIL-C-0081309B(AS) (Pressurized Cans)
MIL-C-0081309C(AS) Type II, Class 1	MIL-C-81309A, Grade A (Bulk) MIL-C-81309A, Grade B (Bulk)
MIL-C-0081309C(AS) Type II, Class 2	MIL-C-81309A, Grade A (Pressurized Cans) MIL-C-81309A, Grade B (Pressurized Cans)

Preparing Activity
Navy(AS)
(DOD Project 8030-N051)

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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